

8.3. Adaptive Control of Linear Systems With Full State Feedback.

- Consider the adaptive control of n^{th} -order linear systems

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_0 y = u \quad (1)$$

- state components $y, \dot{y}, \dots, y^{(n-1)}$ are measurable
- assume coefficient vector $a = \begin{pmatrix} a_n \\ \vdots \\ a_1 \\ a_0 \end{pmatrix}$ is unknown, the sign of a_n is known.

Example: An example of such systems is the dynamics of a mass-spring-damper system

$$m\ddot{y} + c\dot{y} + ky = u$$

Objective: make y closely track the response of a stable reference model

$$a_n y_m^{(n)} + a_{n-1} y_m^{(n-1)} + \dots + a_0 y_m = r(t) \quad (2)$$

with $r(t)$ being a bounded reference signal.

Choice of control law:

- Define a signal $z(t) = y_m^{(n)} - \beta_{n-1} e^{(n-1)} - \dots - \beta_0 e$ (3)

with $\beta_1, \dots, \beta_n$ being positive constants chosen s.t. $p^n + \beta_{n-1} p^{n-1} + \dots + \beta_0$ is a stable (Hurwitz) polynomial. i.e. $1 > \beta_{n-1} > \dots > \beta_0 > 0$.

- Adding $(-a_n z(t))$ to both sides of (1) and rearranging, the plant dynamics as

$$a_n[y^{(n)} - z] = u - a_n z - a_{n-1} y^{(n-1)} - \dots - a_0 y \quad (4)$$

• Choose the control law to be

$$u = \hat{a}_n z + \hat{a}_{n-1} y^{(n-1)} + \dots + \hat{a}_0 y = v^T(t) \hat{a}(t) \quad (5) \quad \text{where } v(t) = \begin{pmatrix} z(t) \\ y^{(n-1)} \\ \vdots \\ \dot{y} \\ y \end{pmatrix}$$

$$\hat{a}(t) = \begin{pmatrix} \hat{a}_n \\ \vdots \\ \hat{a}_0 \end{pmatrix} \rightarrow \text{estimated parameter vector}$$

$\Rightarrow$ The tracking error $e = y - y_m$ satisfies the closed-loop dynamics

$$a_n[e^{(n)} + \beta_{n-1} e^{(n-1)} + \dots + \beta_0 e] = v^T(t) \tilde{a}(t), \quad \text{where } \tilde{a} = \hat{a} - a. \quad (6)$$

[Substituting (3), (5) into (4), then

$$\begin{aligned} & a_n[y^{(n)} - y_m^{(n)} + \beta_{n-1} e^{(n-1)} + \dots + \beta_0 e] \\ &= \hat{a}_n z + \hat{a}_{n-1} y^{(n-1)} + \dots + \hat{a}_0 y - a_n z - a_{n-1} y^{(n-1)} - \dots - a_0 y, \end{aligned}$$

i.e.

$$\begin{aligned} a_n[e^{(n)} + \beta_{n-1} e^{(n-1)} + \dots + \beta_0 e] &= (\hat{a}_n - a_n) z + (\hat{a}_{n-1} - a_{n-1}) y^{(n-1)} + \dots + (\hat{a}_0 - a_0) y \\ &= v^T(t) \tilde{a}(t). \end{aligned}$$

Choice of adaptation law

• Rewrite the closed-loop error dynamics (6) in state space form

$$\dot{x} = Ax + b \left[\left(\frac{1}{a_n} \right) v^T \tilde{a} \right]$$

$$e = cx.$$

where

$$A = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -\beta_0 & -\beta_1 & -\beta_2 & \dots & -\beta_{n-1} \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}, \quad C = (1, 0, \dots, 0).$$

- Consider the Lyapunov function candidate

$$V(X, \tilde{a}) = X^T P X + \tilde{a}^T T^{-1} \tilde{a}$$

where both T and P are symmetric positive definite constant matrices, and P satisfies

$$PA + A^T P = -Q, \quad Q = Q^T > 0 \quad \text{for a chosen } Q$$

- The derivative $\dot{V}$ can be computed as

$$\dot{V} = -X^T Q X + 2\tilde{a}^T v b^T P X + 2\tilde{a}^T T^{-1} \dot{\tilde{a}}$$

$\Rightarrow$ Therefore, the adaptation law $\dot{\tilde{a}} = -T v b^T P X$ leads to

$$\dot{V} = -X^T Q X.$$

$\Rightarrow$ Therefore, with the adaptive controller defined by control law (5) and adaptation law (7), e and its $(n-1)$ derivatives converge to 0.

§8.4. Adaptive Control of Linear Systems with Output Feedback.

- only output measurement, rather than full state feedback.
- A linear time-invariant system can be represented by the transfer function

$$W(p) = K_p \frac{Z_p(p)}{R_p(p)}, \quad \text{where } K_p \text{ is called the high-frequency gain.}$$

$$R_p(p) = a_0 + a_1 p + \dots + a_{n-1} p^{n-1} + p^n$$

$$Z_p(p) = b_0 + b_1 p + \dots + b_{m-1} p^{m-1} + p^m$$

- In our adaptive control problem, the coefficients a_i, b_j , $i = 0, 1, \dots, n-1$, $j = 0, 1, \dots, m-1$

and the high frequency gain k_p are all assumed to be unknown.

- The desired performance is assumed to be described by a reference model with transfer function

$$W_m(p) = K_m \frac{Z_m}{R_m}$$

where Z_m, R_m are monic Hurwitz polynomials of degrees n_m and m_m . k_m is positive.

- In our treatment, we will assume $n_m - m_m \geq n - m$.

Objective The objective of the design is to determine a control law, and an associated adaptation law, so that the plant output y asymptotically approaches y_m .

8.4.1. Linear Systems with Relative Degree One.

When the relative degree is 1, i.e. $m = n - 1$. (The degree of denominator is one order higher than that of numerator)

Example: • Consider the plant described by

$$y = \frac{k_p(p + b_p)}{p^2 + a_{p1}p + a_{p2}} u$$

and the reference model

$$y_m = \frac{k_m(p + b_m)}{p^2 + a_{m1}p + a_{m2}} r$$

- Let the controller with the control law being

$$u = \alpha_1 z + \frac{\beta_1 p + \beta_2}{p + b_m} y + k r$$

where $z = \frac{n}{p + b_m}$, $\alpha_1, \beta_1, \beta_2, k$ are controller parameters.

• If we take these parameters to be

$$\alpha_1 = b_p - b_m,$$

$$\beta_1 = \frac{\alpha_{m1} - \alpha_{p1}}{k_p},$$

$$\beta_2 = \frac{\alpha_{m2} - \alpha_{p2}}{k_p}$$

$$k = \frac{k_m}{k_p}$$

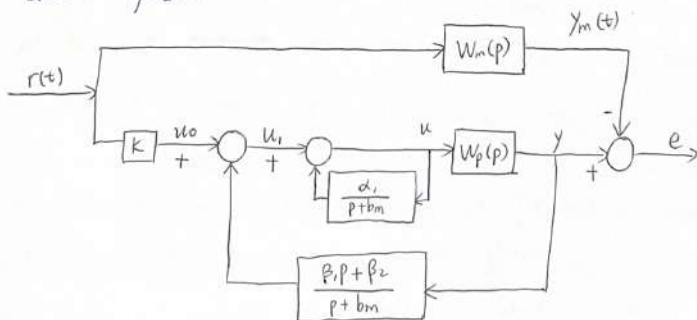
⇒ one can show that the transfer function from the reference input r to the plant output y is

$$W_{ry} = \frac{k_m(p + b_m)}{p^2 + a_m p + \alpha_{m2}} = W_m(p) \quad \text{I have no idea how to calculate this}$$

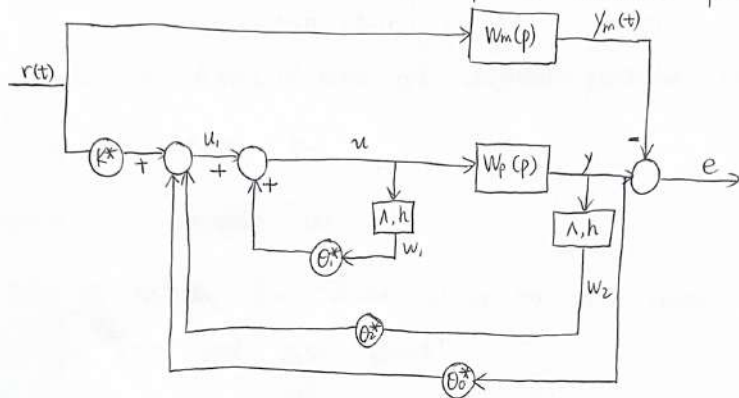
⇒ perfect tracking is achieved with this control law, i.e. $y(t) = y_m(t)$.

(What does this mean?)

Control system:



- The controller structure for second-order plants can be extended to any plant with relative degree one. The resulting structure is the following, where k^* , θ_1^* , θ_2^* and θ_0^* represents controller parameters.



- The block for generating the filter signal w_1 represents an $(n-1)^{th}$ order dynamics,

$$\dot{w}_1 = \Lambda w_1 + h u \quad \text{where } w_1 \text{ is an } (n-1) \times 1 \text{ state vector,}$$

Λ is an $(n-1) \times (n-1)$ matrix, and

h is a constant.

$$\det(pI - \Lambda) = Z_{n-1}(p)$$

- The block for generating the $(n-1) \times 1$ vector w_2 has the same dynamics but with y as input, i.e.

$$\dot{w}_2 = \Lambda w_2 + h y$$

- The scalar gain k^* is defined to be $k^* = \frac{k_m}{k_p}$ and is intended to modulate the high-frequency gain of the control system.
- The vector θ_1^* contains $(n-1)$ parameters which intend to cancel the zeros of the plant.
- The vector θ_2^* contains $(n-1)$ parameters which, together with the scalar

gain θ_0^* can move the poles of the closed-loop control system to the locations of the reference model poles.

$\Rightarrow$ To this end, the control law is chosen to be

$$u = k(t)r + \theta_1(t)w_1 + \theta_2(t)w_2 + \theta_0(t)y, \quad (8)$$

where $k(t)$, $\theta_1(t)$, $\theta_2(t)$, $\theta_0(t)$ are controller parameters to be provided by the adaptation law.

Choice of adaptation law

- Let θ be the 2×1 vector, w be the 2×1 vector, i.e.

$$\theta(t) = [k(t) \quad \theta_1(t) \quad \theta_2(t) \quad \theta_0(t)]^T$$

$$w(t) = [r(t) \quad w_1(t) \quad w_2(t) \quad y(t)]^T$$

Then the control law (8) can be compactly written as

$$u = \theta^T(t)w(t) \quad (9)$$

- Denote the ideal value of θ by θ^*

the error between $\theta(t)$ and θ^* by $\phi(t) = \theta(t) - \theta^*$,

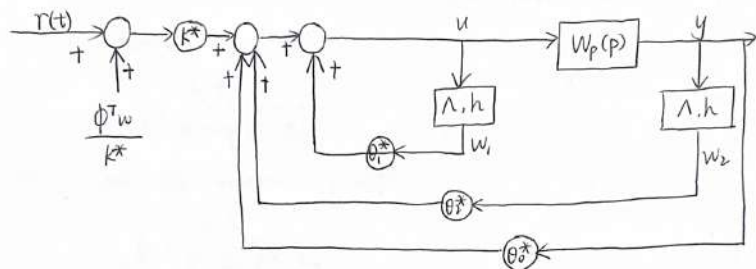
the estimated parameters $\theta(t)$ can be represented as $\theta(t) = \theta^* + \phi(t)$.

Then the control law (8) can be written as

- $$u = (\theta^*)^T w + \phi^T(t) w.$$

- In order to choose an adaptation law so that the tracking error e converges to zero, we have to first find out how the tracking error is related to the parameter error.

- With the control law given in (9), the control system with variable gains can be represented as follows, with $\frac{\phi^T(t)w}{k^*}$ regarded as an external signal.



the output here must be

$$y(t) = W_m(p)r + W_m(p) \cdot \frac{\phi^T(t)w}{k^*}$$

Since $y_m(t) = W_m(p)r$, the tracking error is seen to be related to the parameter error by

$$e(t) = W_m(p) \cdot \frac{\phi^T(t)w(t)}{k^*}$$

- According to lemma 8.1. the following adaptation law is chosen

$$\dot{\theta} = -\text{sgn}(k_p) \gamma e(t) w(t)$$

§ 8.4.2. Linear Systems with Higher ^{relativ} Relative Degree.

- The design of adaptive controller for plants with relative degree larger than 1 is both similar to, and different from, that for plants with relative degree 1.

- Specifically, the choice of control law is quite similar but the choice of adaptation law is very different.

Choice of control law

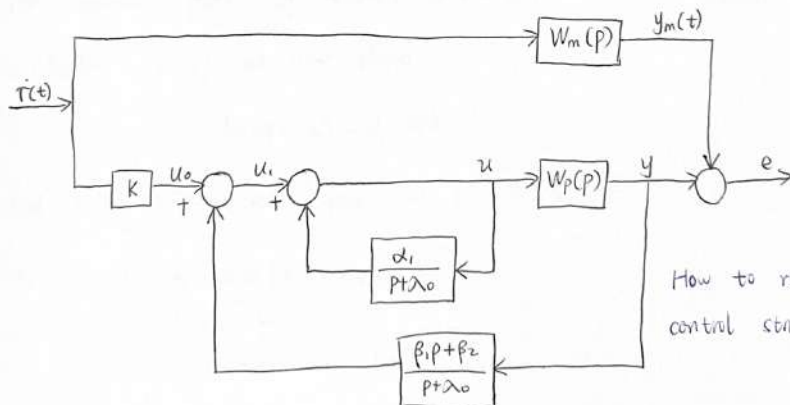
Example: Consider the second order plant described by the transfer function

$$y = \frac{k_p u}{p^2 + a_{p1}p + a_{p2}}$$

and the reference model

$$y_m = \frac{k_m r}{p^2 + a_{m1}p + a_{m2}}$$

- Let us consider the control structure shown as follows.



How to read this control structure?

- The closed-loop transfer function from the reference signal r to the plant output y is

$$W_{ry} = k \frac{\frac{p+\lambda_0}{p+\lambda_0+\alpha_1} \frac{k_p}{p^2+a_{p1}p+a_{p2}}}{1 + \frac{p+\lambda_0}{p+\lambda_0+\alpha_1} \frac{\beta_1 p + \beta_2}{p+\lambda_0} \frac{k_p}{p^2+a_{p1}p+a_{p2}}} = \frac{k k_p (p+\lambda_0)}{(p+\lambda_0+\alpha_1)(p^2+a_{p1}p+a_{p2}) + \frac{k_p(\beta_1 p + \beta_2)}{p+\lambda_0}}$$

$\Rightarrow$ Therefore, if the controller parameters $\alpha_1, \beta_1, \beta_2$ and k are chosen such that

$$(p + \lambda_0 + \alpha_1)(p^2 + \alpha_{p1}p + \alpha_{p2}) + k_p(\beta_1 p + \beta_2) = (p + \lambda_0)(p^2 + \alpha_{m1}p + \alpha_{m2})$$

$$k = \frac{k_m}{k_p},$$

$$\text{then } W_{ry} = \frac{\frac{k_m}{k_p} \cdot k_p \cdot (p + \lambda_0)}{(p + \lambda_0)(p^2 + \alpha_{m1}p + \alpha_{m2})} = \frac{k_m}{p^2 + \alpha_{m1}p + \alpha_{m2}}$$

i.e. the closed-loop transfer function W_{ry} becomes identically the same as that of the reference model. Clearly, such choice of parameters exists and is unique.

- For general plants of relative degree larger than 1, instead of $\det[pI - A] = Z_m(p)$, we now choose

$$\det[pI - A] = Z_m(p)\lambda_1(p)$$

and $\lambda_1(p)$ is a Hurwitz polynomial of degree $(n-1-m)$.

- The closed-loop transfer function is

$$W_{ry} = \frac{k k_p z_p \lambda_1(p) Z_m(p)}{R_p(p)[\lambda(p) + c(p)] + k_p z_p D(p)}$$

Now, the question is whether in this general case there exist choice of values for k, θ_0, θ_1 and θ_2 such that the above transfer function becomes exactly the same as $W_m(p)$, or equivalently,

$$R_p(\lambda(p) + c(p)) + k_p z_p D(p) = \lambda_1 z_p R_m(p). \quad (10)$$

- The answer to this question can be obtained from the following

Lemma: Let $A(p)$ and $B(p)$ be polynomials of degree n_1 and n_2 .

If $A(p)$ and $B(p)$ are relatively prime, then there exist polynomials $M(p)$ and $N(p)$ such that

$$\underline{A(p)} M(p) + \underline{B(p)} N(p) = \underline{A^*(p)},$$

where $A^*(p)$ is an arbitrary polynomial.

- This lemma can be used straightforwardly to answer our question regarding (10). $\underline{P_p} (\lambda(p) + C(p)) + \underline{k_p Z_p} D(p) = \underline{\lambda_1 Z_p R_m(p)}$

By regarding P_p as $A(p)$, $k_p Z_p$ as $B(p)$ and $\lambda_1 Z_p R_m$ as $A^*(p)$, we conclude that there exist polynomials $\lambda(p) + C(p)$ and $D(p)$ s.t. (10) is satisfied.

Choice of Adaptation law

- We again use a control law of the form

$$u = \theta^T(t) w(t)$$

Using a similar reasoning as before, we can again obtain the output y in the form of $y(t) = W_m(p)r + W_m(p)[\phi^T(t)w/k^*]$ and the tracking error is in the form of

$$e(t) = W_m(p)[\phi^T w/k^*] \quad (11)$$

- However, the choice of adaptation law given by $\dot{\theta} = -\text{sgn}(k_p) r e(t) w(t)$ cannot be used. (Unlike to the case of plant with relative degree 1) because now the reference model transfer function $W_m(p)$ is no longer SPR.

- A famous technique called error augmentation ^{误差增强} can be used to avoid the difficulty in finding an adaptation law for $e(t) = W_m(p) [\phi^T(t) w] / k^*$.
- Specifically, let us define an auxiliary error $\eta(t)$ by
 ^{辅助误差}

$$\eta(t) = \theta^T(t) W_m(p) [w] - W_m(p) [\theta^T(t) w(t)] \quad (12)$$

and define an augmented error $\varepsilon(t)$ by combining the tracking error $e(t)$ with the auxiliary error $\eta(t)$ as

$$\varepsilon(t) = e(t) + \alpha(t) \eta(t) \quad (13)$$

~~For~~ Substituting (10), (12) into (13), we obtain

$$\varepsilon(t) = \frac{1}{k^*} \phi^T(t) \bar{w} + \phi_\alpha \eta(t),$$

$$\text{where } \phi_\alpha = \alpha(t) - \frac{1}{k^*}, \quad \bar{w}(t) = W_m(p) [w].$$

- A number of standard techniques to be discussed in §8.7 can be used to demonstrate that the controller parameters $\theta(t)$ and the parameter $\alpha(t)$ for forming the augmented error are updated by

$$\dot{\theta} = - \frac{\text{sgn}(k_p) \gamma \varepsilon \bar{w}}{1 + \bar{w}^T \bar{w}}, \quad \dot{\alpha} = - \frac{\gamma \varepsilon \eta}{1 + \bar{w}^T \bar{w}} \quad (14)$$

- With the control law $u = \theta^T(t) w(t)$ and adaptation law (14), global convergence of the tracking error can be shown.

§ 8.5. Adaptive Control of Nonlinear Systems.

Problem Statement We consider n^{th} -order nonlinear systems

$$y^{(n)} + \sum_{i=1}^{n-1} d_i f_i(x, t) = bu, \quad (15)$$

where $x = [y, \dot{y}, \dots, y^{(n-1)}]^T$ is the state vector

f_i are known nonlinear functions of the state and time.

and the parameters α_i and b are unknown constants. We assume that the state is measured, and that the sign of b is known. One example of such dynamics is

$$m\ddot{x} + c f_1(\dot{x}) + k f_2(x) = u,$$

which represents a mass-spring-damper system with nonlinear friction and nonlinear damping.

- The objective of the adaptive control design to make the output asymptotically tracks a desired output $y_d(t)$ despite the parameter uncertainty
- Let us rewrite (15) as

$$h y^{(n)} + \sum_{i=1}^n \alpha_i f_i(x, t) = u, \quad (16)$$

by dividing both sides by the unknown constant b , where $h = \frac{1}{b}$ and $\alpha_i = \frac{\alpha_i}{b}$

choice of control law

- define a combined error

$$s = e^{(n-1)} + \lambda_{n-2} e^{(n-2)} + \dots + \lambda_0 e \quad \text{~~define~~}$$

Note that s can be rewritten as

$$s = y^{(n-1)} - y_r^{(n-1)},$$

↓

$$y_d^{(n-1)} - \lambda_{n-2} e^{(n-2)} - \dots - \lambda_0 e.$$

- Consider the control law

$$u = h y_r^{(n)} - k s + \sum_{i=1}^n \alpha_i f_i(x, t), \quad (17)$$

where k is a constant of the same sign as h ,

$$y_r^{(n)} = y_d^{(n)} - \lambda_{n-2} e^{(n-2)} - \dots - \lambda_0 \dot{e}$$

- Substituting (17) into (16), we see

$$h\dot{y}_r^{(n)} + \sum_{i=1}^n a_i \cancel{f_i}(x, t) = h\dot{y}_r^{(n)} - ks + \sum_{i=1}^n \cancel{a_i} \cancel{f_i}(x, t)$$

$$h\dot{s} + ks = 0$$

and therefore gives exponential convergence of s , which, in turn, guarantees the convergence of e .

Choice of adaptation law

- For our adaptive control, the control law (17) is replaced by

$$u = \hat{h} \dot{y}_r^{(n)} - ks + \sum_{i=1}^n \hat{a}_i f_i(x, t),$$

where h and the a_i have been replaced by their estimated values. The tracking error from this control law can be easily shown to be

$$h\dot{s} + ks = \tilde{h} \dot{y}_r^{(n)} + \sum_{i=1}^n \tilde{a}_i f_i(x, t).$$

- This can be rewritten as

$$s = \frac{1/h}{p_1(k/h)} \left[\tilde{h} \dot{y}_r^{(n)} + \sum_{i=1}^n \tilde{a}_i f_i(x, t) \right].$$

Lemma 8.1. suggests us to choose the following adaptation law

$$\dot{\hat{h}} = -\gamma \operatorname{sgn}(h) s \dot{y}_r^{(n)}$$

$$\dot{\hat{a}}_i = -\gamma \operatorname{sgn}(h) s f_i$$